Lecture 6: Consumer Demand

Live Session

Purpose of this live session

- **Recap** the pre-recorded lectures

- **Discuss** application of consumer demand theory to data
  - Double log specification
  - Budget share specification

Recap of the pre-recorded lectures：
Consumer demand theory

Determinants of consumer demand

- Suppose there are J goods in the market
- The household/individual decides how much of each good to purchase in order to maximize utility
  - We denote $(q_1, q_2, \ldots, q_J)$ the quantities demanded of good $1, 2, \ldots, J$, respectively
- Demand depends on:
  - Budget constraint
    - Income y (= total expenditures if no saving/borrowing)
    - Prices of goods $p_1, p_2, \ldots, p_J$
  - Preferences: tastes, age, household characteristics

Effects of income changes on demand

- Consumer theory does not constrain the direction in which an increase in income affects the demand for a single good
- If the demand rises with income, the good is a **normal good**
- If the demand falls with income, the good is an **inferior good**
- Two types of normal goods:
  1. **Necessity goods**: Budget share *decreases* as income rises
  2. **Luxury goods**: Budget share *increases* as income rises

Income elasticity

- Income elasticities can be interpreted to characterise which type of goods we're dealing with

Characterising goods by their income elasticities

- A good is inferior if $\xi_{q_j y} < 0$
- A good is normal if $\xi_{q_j y} > 0$
- A normal good is a necessity if $0 < \xi_{q_j y} < 1$
- A normal good is a luxury if $\xi_{q_j y} > 1$

Price elasticity of demand

- Goods can be divided into those for which total spending (and therefore budget share) rises and those for which it falls as price rises

Characterising goods by their price elasticities

- Good j is **Giffen** if: $\xi_{q_j p_j} > 0$
- Good j is **price inelastic** if $-1 < \xi_{q_j p_j} < 0$
  - Total spending (and therefore budget share) rises as price rises
- Good j is **unitary elastic** if $\xi_{q_j p_j} = -1$
  - Total spending (and therefore budget share) stays constant as price rises
- Good j is **price elastic** if $\xi_{q_j p_j} < -1$
  - Total spending (and therefore budget share) falls as price rises

Cross-price elasticities

- Cross-price effects can be summarised using cross-price elasticities of demand

Cross-price elasticity of demand

The elasticity of demand for good j with respect to the price of good k is the proportionate response in the quantity demanded for good j to a proportionate change in theprice of good k.

$$\xi_{q_j p_k} = \frac{\partial q_j}{\partial p_k} \cdot \frac{p_k}{q_j} = \frac{\partial \ln q_j}{\partial \ln p_k} \tag{2}$$

where p_k is the price of good k

- Two goods are substitutes if $\xi_{q_j p_k} > 0$
- Two goods are complements if $\xi_{q_j p_k} < 0$

Demographic variables

- In addition to income and prices, consumer demand is affected by preferences
- It is reasonable to think that preferences are affected by demographic characteristics
  - Age, gender, geographical residence, ethnicity, etc.
- Household size can also affect the demand
  - An increase in household size increases a households' needs. For a given budget, it leaves the household's economic standard of living lower.
  - In this respect, it is similar to a negative income shock and therefore affects the demand
- The type of goods bought may also depend on the family composition
  - Adults vs. children
  - Males vs. females

In this live session:

Application of Consumer Demand Theory to Data

Application of consumer demand theory to data

- Theory sometimes restricts the direction (sign) of elasticities
- But to know the magnitude of elasticities, we need to turn to the data
- Why is it useful to know the empirical magnitude of elasticities?
  - Profit maximizing retailer need to know own price elasticities
  - Government using excise duties to restrict demand for alcohol needs to know own price elasticities, and possibly cross price elasticity with other alcohols
  - Government using income supplements to help families buy necessities needs to know income elasticities
- We could also be interested in how demographic characteristics affect the demand for certain goods

Regression model

- Estimation requires a choice for the functional form linking quantities demanded to the explanatory variables (price, income, characteristics...)
- In choosing the functional form, we want to:
  - Maintain ease of interpretation
  - Keep flexibility in describing the data
  - Allow restrictions suggested by consumer theory to be tested
- We will consider two examples of specification:
  1. Double log demand equation
  2. Budget share specification

Double log demand function

- The double-log demand function has the following form:

$$\ln q_{ij} = \alpha + \beta \ln y_i + \sum_{k=1}^{J} \gamma_k \ln p_{ik} + \delta Z_i + \epsilon_i \tag{3}$$

  - q_{ij} is quantity of good j consumed by household i
  - y_i is total income (= total expenditure or budget) of household i
  - p_{ik} is price paid by household i for good k (notice that the sum is over all the goods, including good j)
  - Z_i is a vector of demographic characteristics for household i
  - ϵ_i is the error term
- This functional form is also known as the **loglinear** or **constant elasticity** specification
  - Note that it corresponds to the log-log functional form we saw in Lecture 4

Interpretation of the regression parameters

- This specification has the advantage that the parameters can be directly interpreted as elasticities
- β is the income elasticity of demand for good j:

$$\xi_{q_j y} = \frac{\partial q_{ij}}{\partial y_i} \frac{y_i}{q_{ij}} = \frac{\partial \ln q_{ij}}{\partial \ln y_i} = \beta \tag{4}$$

- γ_j is the own price elasticity of demand for good j

$$\xi_{q_j p_j} = \frac{\partial q_{ij}}{\partial p_{ij}} \frac{p_{ij}}{q_{ij}} = \frac{\partial \ln q_{ij}}{\partial \ln p_{ij}} = \gamma_j \tag{5}$$

- γ_k $(k \neq j)$ is the cross price elasticity for good j with respect to price of good k

$$\xi_{q_j p_k} = \frac{\partial q_{ij}}{\partial p_{ik}} \frac{p_{ik}}{q_{ij}} = \frac{\partial \ln q_{ij}}{\partial \ln p_{ik}} = \gamma_k \tag{6}$$

Interpretation of the regression parameters

- Income elasticity:
  - Good j is inferior if $\beta < 0$
  - It is a normal good and a necessity if $0 < \beta < 1$
  - It is a normal good and a luxury if $\beta > 1$
- Own price elasticity:
  - Good j if a Giffen good if $\gamma_j > 0$
  - It is price inelastic if $-1 < \gamma_j < 0$
  - It is price elastic if $\gamma_j < -1$
- Cross price elasticity
  - Good j is a substitute for good k if $\gamma_k > 0$
  - Good j is a complement for good k if $\gamma_k < 0$ where $k \neq j$

Remarks on the double log demand equation

- The double log specification has the advantage that its parameters are easy to interpret as elasticities
- But it has several disadvantages:
  - It cannot be used for households purchasing none of the good since the dependent variable would be undefined
  - It imposes constant elasticities, which may not be a reasonable assumption

$\Rightarrow$ We look at another specification, the budget share specification, which does not have any of these problems

Budget share specification

- The budget share specification relates the budget share linearly to the same explanatory variables as in the double log specification
- The budget share for good j is the proportion of total budget spent on good j by household i:

$$w_{ij} = \frac{p_{ij} q_{ij}}{y_i} \tag{7}$$

- The budget share specification can be written as:

$$w_{ij} = \alpha + \beta \ln y_i + \sum_{k=1}^{J} \gamma_k \ln p_{ik} + \delta Z_i + \epsilon_i \tag{8}$$

Interpretation of the regression parameters

- **In the budget share specification, the interpretation of the coefficients is different than it is in the double log specification**
- The total budget (income) elasticity is no longer β, but:

$$\frac{\partial q_{ij}}{\partial y_i} \cdot \frac{y_i}{q_{ij}} = \frac{\partial \ln q_{ij}}{\partial \ln y_i} = 1 + \frac{\beta}{w_{ij}} \tag{9}$$

- This implies that:
  - Good j is a luxury good if $\beta > 0$ (i.e. the budget share rises as total budget rises)
  - Good j is a necessity if $\beta < 0$

Budget share specification

- The coefficients on the log price terms are not to be interpreted as elasticities either
- However there are still simple rules of interpretation
- The budget share rises with own price and the good is therefore price inelastic if $\gamma_j > 0$
- It is price elastic if $\gamma_j < 0$
- The good is a substitute for the k-th good if $\gamma_k > 0$ and a complement for the k-th good if $\gamma_k < 0$ where $k \neq j$

Application of Consumer Demand Theory to Data: Examples

Example 1: The demand for eggs

- We use cross-sectional data on the demand for eggs in Czech Republic
- The dataset contains information on
  - Number of eggs purchased by households
  - Prices faced by households
  - Demographic characteristics (age of household head, number of children, number of adults)

The demand for eggs: Double log specification

$$\begin{aligned} \ln eggs_i &= \alpha + \beta \ln exp_i + \gamma_1 \ln price_eggs_i + \gamma_2 \ln price_bread_i \\ &+ \delta_1 age_i + \delta_2 kids_i + \delta_3 adults_i + \epsilon_i \end{aligned}$$

where exp is total household expenditure.

Number of obs = 1241
R^2=0.126
St error=1.070

Dependent variable: *ln quantity eggs*

	Coefficient	***t* value**
ln expenditure	0.102	1.246
ln price eggs	-1.848	-10.875
ln price bread	0.623	2.651
age	0.009	3.017
no of children	-0.0001	-0.263
no of adults	0.003	3.449
Constant	-0.302	-0.243

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Multiple Choice

Based on the estimated coefficient, are eggs:

- necessity goods
- inferior goods
- luxury goods

Multiple Choice

Based on estiamted coefficient, are eggs

- price inelastic
- price elastic
- giffen goods

The demand for bread: Double log specification

$$\begin{aligned} \ln bread_i &= \alpha + \beta \ln exp_i + \gamma_1 \ln price_eggs_i + \gamma_2 \ln price_bread_i \\ &+ \delta_1 age_i + \delta_2 kids_i + \delta_3 adults_i + \epsilon_i \end{aligned}$$

Number of obs = 1241
R^2=0.460
St error=0.463

Dependent variable: *ln quantity bread*

	Coefficient	***t* value**
ln expenditure	0.164	4.645
ln price eggs	0.014	0.185
ln price bread	-0.938	-9.222
age	0.008	6.601
no of children	0.002	13.943
no of adults	0.005	13.297
Constant	0.856	1.589

Multiple Choice

Across the two demand equatiions, are the symmetry restrictions satisfied?

- yes
- no

The demand for eggs: Budget share specification

$$
\begin{aligned}
w_i &= \alpha + \beta \ln exp_i + \gamma_1 \ln price_eggs_i + \gamma_2 \ln price_bread_i \\
&+ \delta_1 age_i + \delta_2 kids_i + \delta_3 adults_i + \epsilon_i
\end{aligned}
$$

where w_i is the share of household i's total budget spent on eggs

Number of obs = 1241
R^2=0.368
St error=0.0043

Dependent variable: *eggs share*

	Coefficient	***t* value**
ln expenditure	-.00545	-16.612
ln price eggs	-.00093	-1.366
ln price bread	.00073	0.772
age	.00007	6.536
no of children	0.000004	2.332
no of adults	.00002	5.145
Constant	.04574	9.140

- Average budget share for eggs is 0.00607 so total budget elasticity is 1 - 0.00545/0.00607 = 0.102.

Multiple Choice

Based on estimated coefficient, what is the relationship between eggs and bread?

- substitutes
- complements

The demand for eggs: Interpretation concerns

- Eggs are not entirely homogeneous good: they differ in shape and quality etc.
- Suppose individuals with a strong preference for eggs also tend to choose higher-quality (and therefore more expensive) eggs.
- In that case, quality is positively correlated with both price and quantity demanded.
- However, quality is not included in the regression → omitted variable problem.
- Because the omitted variable (quality) is positively correlated with both the outcome and the price, the estimated price coefficient will be biased upward.
- More generally, any unobserved factor correlated with both price and demand can generate omitted variable bias when estimating price effects from a single cross-section.

Example 2

You are given data on food expenditure, income and demographic characteristics of 150 married couples, where the husband is a manual worker and is aged 30-50. All survey interviews take place in the same month.

You decide to estimate an Engel curve and regress the ratio of food expenditure to income (*wfoy*) on the logarithm of net current income (*ncy*), the number of children aged 6-10 (*kids2*), the number of children aged11-17(*kids3*) and an indicator for London(*london*=1 if the couple lives in Greater London, and 0 otherwise).

Your estimates are:

OLS Estimation, Dependent variable is *wfoy*, 150 observations

Variable	Coefficient	Standard error
constant	0.877	0.065
ncy	-0.155	0.014
kids2	0.026	0.006
kids3	0.138	0.032
london	0.047	0.018
R^2	0.496	
Mean of *wfoy*	0.189	
Mean of *ncy*	4.473	

Short Answer

(a) On the basis of the above estimates, establish if food is a necessity or a luxury.

Short Answer

(b) Establish the statistical significance of the coefficient on london.

Short Answer

(c) Further, compute the income elasticity for a hypothetical couple living outside London, with mean log income and no children.

Example 3: Estimating the gasoline demand elasticity (Lin and Zeng, 2013)

- On January 1, 2009, China initiated a modest reform on its fuel tax, which led to an increase in the gasoline consumption tax.
- Rising gasoline consumption has led to concerns such as local pollution, global warming, and various other issues.
- In this paper, the authors estimate the price and income elasticities of demand for gasoline in China, which is important for evaluating the effectiveness of China's tax reform in decreasing gasoline consumption.

Estimating the gasoline demand elasticity

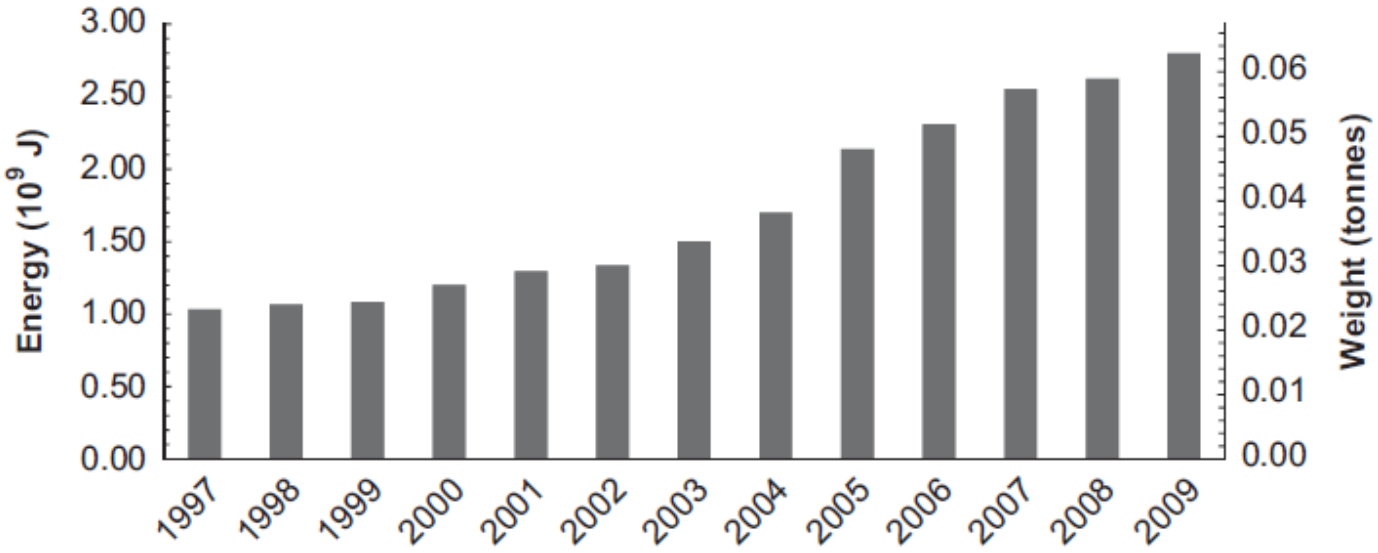

Fig. 3. Annual gasoline consumption per capita, 1997–2009.
Source: China Energy Statistical Year Book, 1997–2009.
Note: Tibet is not included.

Example 2: Estimating the gasoline demand elasticity (Lin and Zeng, 2013)

To estimate the gasoline price elasticity, we start with a basic double log model:

$$\ln D_{it} = \beta_0 + \beta_1 \ln P_{it} + \beta_2 \ln Y_{it} + \varepsilon_{it} \ , \qquad (1)$$

where D_{it} is per capita gasoline demand in gallons for region *i in* year *t*, P_{it} is the real price of gasoline in 2005 constant Yuan for region *i* in year *t*, Y_{it} is real per capita disposable income in 2005 constant Yuan for region *i* in year *t*, and ε_{it} is a mean zero error term. Because of data limitations, the majority of this analysis is carried out using annual data.

Table 3
Double log model of gasoline demand.

	Dependent variable is log per capita gasoline demand		
	OLS (without instruments)	Diesel price as IV	Crude oil price as IV
Log real gasoline price	−0.264**	−0.23**	−0.432***
	(0.088)	(0.090)	(0.097)
Log real income	1.01***	1.01***	1.05***
	(0.038)	(0.038)	(0.039)
Constant	−13.60***	−13.89***	−12.64***
	(0.644)	(0.661)	(0.684)
Observations	275	263	275
R-squared	0.76	0.77	0.76

Notes: Standard errors in parentheses.
*Significant at 5%.
** Significant at 1%.
*** Significant at 0.1%.

Short Answer

What are the estimated price and income elasticities with OLS estimation?

Model with other control variables

$$\ln D_{it} = \beta_0 + \beta_1 \ln P_{it} + \beta_2 \ln Y_{it} + \beta_3 \ln RI_t + \beta_4 UE_t + \varepsilon_{it}.$$

where RI is the real interest rate and UE is the unemployment rate.

Table 5
Double log model of gasoline demand with macroeconomic variables.

	Dependent variable is log per capita gasoline demand		
	(1)	(2)	(3)
Log real gasoline price	−0.196*	0.078	0.057
	(0.089)	(0.109)	(0.111)
Log real income	1.042***	1.110***	1.106***
	(0.038)	(0.040)	(0.040)
Log real interest rate	0.194***		0.071
	(0.054)*		(0.062)
Log unemployment rate		−1.197***	−1.040***
		(0.233)	(0.273)
Constant	−14.750***	−16.131***	−16.180***
	(0.699)	(0.787)	(0.786)
Observations	263	263	263
R-squared	0.790	0.790	0.790

Notes: Standard errors in parentheses.
**Significant at 1%.
* Significant at 5%.
*** Significant at 0.1%.

Table 8
Review of gasoline demand elasticity estimates for developing countries from earlier studies.

Country	Estimation period	*Price elasticity*			*Income elasticity*			Source
		Short run	Intermediate run	Long run	Short run	Intermediate run	Long run	
Hong Kong	1973–1987	0.055			0.22			McRae (1994)
India	1973–1987	−0.32			1.38			McRae (1994)
Indonesia	1973–1987	−0.20			1.69			McRae (1994)
South Korea	1973–1987	−0.50			0.72			McRae (1994)
Malaysia	1973–1987	−0.13			0.57			McRae (1994)
Pakistan	1973–1987	0.39			2.91			McRae (1994)
Philippines	1973–1987	−0.39			0.15			McRae (1994)
Sri Lanka	1973–1987	−0.34			0.82			McRae (1994)
Taiwan	1973–1987	0.024			0.81			McRae (1994)
Thailand	1973–1987	−0.30			1.77			McRae (1994)
Bangladesh	1973–1987	−0.35			0.016			McRae (1994)
India	1972–1994	−0.21		−0.32	1.18		2.68	Ramanathan (1999)
Kuwait	1970–1989	−0.37			0.47			
China	1980–1999	−0.19		−0.56	1.64		0.97	Cheng and Thomson (2004)
China	1997–2008		−0.497 to −0.196			1.01–1.05		This paper

- Next Lecture

Lecture 7: Consumption and Saving